RUBY (RUISHAN) ZHANG

rzhang15@g.harvard.edu
Cell 312-843-2679
rzhang15.github.io

HARVARD UNIVERSITY

Littauer Center 315
1805 Cambridge St
Cambridge MA 02138

Placement Director: Myrto Kalouptsidi	myrto@fas.harvard.edu	617-495-2144
Placement Director: Oleg Itskhoki	itskhoki@fas.harvard.edu	617-495-2144
Administrative Director: Brenda Piquet	bpiquet@harvard.edu	617-495-8927

Education

Harvard University
Ph.D. Economics, 2021 to 2027 (expected)

University of Chicago
B.A. in Economics (with Honors), 2019
B.S. in Mathematics (with Specialization in Economics), 2019
B.S. in Statistics, 2019

Fields

Innovation, Digitization & AI, Industrial Organization

References

Shane Greenstein (Chair)
sgreenstein@hbs.edu

Robin Lee
robinlee@fas.harvard.edu

Ariel Pakes
apakes@fas.harvard.edu

Jesse Shapiro
jesse_shapiro@fas.harvard.edu

Fellowships & Awards

Dissertation Completion Fellowship, Harvard University, 2026
Humane Studies Fellowship, Institute for Humane Studies, 2026
John and Kiendl Gordon Funding Award, Harvard University, 2026
Lab for Economic Applications and Policy Funding Award, Harvard University, 2025-2026
Becker-Friedman Institute Microeconomics Academic Award, University of Chicago, 2019
David S. Hu Thesis Award, University of Chicago, 2019
Phi Beta Kappa (Junior Year), University of Chicago, 2018
Student Marshal, University of Chicago, 2018
Dougan Scholar Certificate Program, University of Chicago Booth School of Business, 2016-2019

Teaching

Teaching Fellow for Professor Jesse Shapiro, *Computing for Economists* (Graduate), 2024 & 2025
Teaching Fellow for Professor Robin Lee, *Industrial Organization* (Undergraduate), 2024 & 2025

Job Market Paper

Returns to Science of Centrally Provisioned GPUs and Optimal Allocation (with Connie Xu)

Artificial intelligence and deep learning methods require expensive capital investments in the form of Graphical Processing Units (GPUs). Given the importance of GPU clusters to computational research, where should infrastructure funding be allocated? We construct a novel panel of university GPU capacity and research output for R1 U.S. universities, combining historical snapshots of research computing websites with bibliometric data from OpenAlex. We use large, discrete computing capacity jumps in an event-study design and find modest effects of cluster upgrades on AI-related publications. However, there is significant heterogeneity: low-tier universities exhibit a larger marginal response to capacity upgrades than high-tier universities. We develop a structural model of the university as a multi-product organization to estimate the production function for GPU-enabled research, derive a capital allocation rule across fields, and use the estimated elasticities to characterize the NSF's revealed preference over efficiency versus equity. Our findings suggest that diminishing returns to computing capital favor a more redistributive allocation of federal cyberinfrastructure grants.

Publications

Greenstein, N., Fontana, R., Greenstein, S., Zhang, R., Zhang, H., Olivarez, O., and Kim, D.Y. Wireless communications equipment markets: evolution, classification, and measurement. *Telecommunications Policy* 50(6), 103209 (2026). https://doi.org/10.1016/j.telpol.2026.103209

Zhang, R. "Building Up Research Capacity at Minority Institutions." Commissioned paper for *A Plan to Promote Defense Research at Minority-Serving Institutions* (2024). https://nap.nationalacademies.org/resource/27838/Zhang_Commissioned_Paper.pdf

Chetty, R., Jackson, M.O., Kuchler, T. *et al.* Social capital I: measurement and associations with economic mobility. *Nature* **608**, 108–121 (2022). https://doi.org/10.1038/s41586-022-04996-4

Chetty, R., Jackson, M.O., Kuchler, T. *et al.* Social capital II: determinants of economic connectedness. *Nature* **608**, 122–134 (2022). https://doi.org/10.1038/s41586-022-04997-3

Working Papers

Scientific Input Costs in University Life Sciences Research (with Connie Xu)

Public funders delegate money and broad discretion to principal investigators (PIs), yet whether that discretion protects research output when input prices rise is unknown. We ask how PIs adjust, and how much research is lost. Using novel purchase-level procurement records, we exploit the 2014 merger of Thermo Fisher Scientific and Life Technologies as a shock to input prices. A matched difference-in-differences design estimates that prices in affected markets rose approximately 20 percent while quantities moved minimally, indicating highly inelastic demand. Tracing this cost shock through a panel of 6,382 U.S. life sciences PIs, we find the average PI, facing a 3.2 percent rise in input costs, publishes about 1.2 percent fewer papers, an elasticity of about -0.38. The decline concentrates among early-career PIs, who command fewer resources to draw on. The volume of publicly funded science thus depends on prices in input markets and on the PIs who manage the grants.

Secrecy, Spectrum, and Certification: Evidence from Wireless Electronic Devices
(with Roberto Fontana, Shane Greenstein, and Do Yoon Kim)

This paper examines the strategic decisions of firms to withhold information in innovation-intensive markets, highlighting that secrecy contains distinct temporal dimensions. Using all FCC equipment filings from 2001 to 2021, we analyze firms' choices between long-term confidentiality (LTC), which permanently conceals technical information such as block diagrams, and short-term confidentiality (STC), which temporarily delays the public release of marketing information to ensure against launch and coordination risks in the commercialization process. LTC use reflects appropriability concerns and institutional complexities—technically complex products and competitive markets are 12 to 35% more likely to adopt it. By contrast, STC is driven by the commercialization timing of branded products by large firms. Large, domestic, patent-owning firms with regulatorily complex products are more likely to use STC. Technical aspects of the product do not predict STC usage. An exposure-based event study further shows that firms with a high ex-ante propensity to use STC bring products to market faster after the policy becomes available, with no corresponding changes in product complexity or volume.

Conferences

IIOC, DRUID, AOM, EARIE, DISS, TPRC (2026)

Research Positions

Research Assistant to Shane Greenstein, Harvard Business School, 2022-2025
Research Assistant to Raj Chetty and Nathaniel Hendren, Harvard University, 2019-2021
Research Assistant to James Heckman, University of Chicago, 2016-2018

Academic Service

Organizer, Harvard Industrial Organization Reading Group, 2024-2026
Mentor, MIT and Harvard Economics Application Assistance and Mentoring Program, 2022-2023

Languages

English (native), Mandarin Chinese (fluent), French (basic)

Citizenship

Canada